Patient management guidelines with BPaLM/BPaL: 2024

A supplement to PMDT guidelines - 2021

National Programme for Tuberculosis Control and Chest Diseases (NPTCCD)

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PREFACE

Tuberculosis (TB), and in particular drug-resistant TB (DR-TB), continues to be a major public health challenge globally and in Sri Lanka. In response to evolving scientific evidence at global level, the National Programme for Tuberculosis Control and Chest Diseases (NPTCCD) is committed to ensure that our national protocols are aligned with international best practices while being locally relevant.

The *Programmatic Management of Drug-Resistant TB (PMDT) Guidelines* were last comprehensively updated in 2021. Since then, significant advances have occurred, particularly regarding shorter, all-oral regimens for the treatment of MDR/RR-TB. Following pivotal clinical trials such as the Nix-TB, ZeNix, and TB-PRACTECAL, the World Health Organization has endorsed the use of shorter regimens such as BPaL and BPaLM. These regimens not only demonstrate high efficacy and safety profiles but also provide an opportunity to improve treatment adherence and patient outcomes.

This 2024 supplement to the PMDT Guidelines is intended to provide practical guidance on the implementation of BPaLM and BPaL regimens within the national program. It also briefs on other WHO shorter treatment regimen options. It outlines critical updates on diagnosis, treatment initiation, monitoring, adverse event management, and patient outcome evaluation, all of which are essential for the effective programmatic rollout of these newer regimens.

The development of this guideline supplement was made possible through the contributions of national experts, field practitioners, and international partners. I extend my gratitude to all stakeholders who shared their expertise and time, especially the editorial and technical panels, for their rigorous efforts in reviewing and compiling this document. I hope that this supplement will serve as a valuable resource for clinicians, programme managers, and healthcare workers involved in the management of DR-TB, contributing towards achieving our national and global goals for TB elimination.

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ABBREVIATIONS

Abbreviation	Name
ATP	Adenosine triphosphate
ALLR	All oral longer regimen
Bdq	Bedaquiline
cfu	Colony forming units
Dlm	Delamanid
E	Ethambutol
Eto	Ethionamide
H	Isoniazid
LoD	Level of Detection
Lfx	Levofloxacin
Lzd	Linezolid
Mfx	Moxifloxacillin
OSSTR	Oral Standard Short-course Treatment Regimen
Pa	Pretomanid
PMDT	Programmatic Management of Drug-Resistant TB
Z	Pyrazinamide
R	Rifampicin
S	Streptomycin
WRD	WHO recommended rapid diagnostics
SRLN	WHO supranational laboratory
FQ	fluoroquinolone

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Chapter 1: INTRODUCTION

1.1 Background

The latest Programmatic Management of Drug-Resistant TB (PMDT) guidelines for Sri Lanka was updated during 2021 and currently being practiced. The guideline was updated based on the WHO guidelines on PMDT which was published in 2020. The new PMDT guideline highlights three main treatment regimens for MDR patients; All oral longer regimen (ALLR), Oral Standard Short-course Treatment Regimen (OSSTR) and BPaL. Sri Lanka has converted to all oral drug regimen since 2019 and in this guideline, it is clearly outlined as all oral drugs for standard longer treatment regimen. For the first time, two new regimens were introduced in this guideline; OSSTR and BPaL. OSSTR has a shorter duration of treatment which is 9-11 months. In addition, a further treatment regimen of 6-9 months is introduced consisting of Bedaquiline(Bdq), Pretomanid(Pa) and Linezolid(Lzd).

However, within the recent years, there had been rapid knowledge development in the field, leading to modification of the existing regimen to be more efficient while user friendly. Shorter duration as well as side effects against efficiency was considered in deciding on newer regimen. Based on three major studies Nix TB Trial, ZeNix Trial and TB Practecal Trial, existing BPaL regimen was modified and a newer BPaLM treatment regimen was introduced. This was published as a rapid communication by WHO; DRTB treatment 2022 update.

According to this guideline following are introduced;

1. 6 month BPaLM regimen (Bdq, Pa, Lzd(600mg) and Moxifloxacillin (Mfx) to be used programmatically in place of 9 month or longer (>18 months) regimens, in patients14years and above with MDR/RR TB And BPaL (6-9 months) for RR/MDR TB with FQ resistance.

2. 9 month all oral Bdq regimen are preferred over the longer regimen in children (excluding <6 years, <16kg) and adults with MDR/RR TB. (In WHO 2025 update, this is recommended for all children).

3. Longer regimen preferred for patients with indications and for extensive forms of DRTB (e.g., XDR TB)

Further changes are noted in patient registration and treatment outcomes, which are now aligned with Drug Sensitive TB. With the BPaLM shorter regimen, the recording and reporting formats also are modified. The above changes are discussed in detail under the relevant sections.

1.2 Epidemiology

The estimates of MDR/RR TB for Sri Lanka is around 86 patients per year. The estimated % of new TB cases previously treated TB cases with MDR/RR TB is 0.46% and 3% respectively. The rate of reporting of MDR/RR TB cases is 0.37/100, 000 population (Global TB Report, 2023). However, the notified MDR/RR TB patients remain well below that.

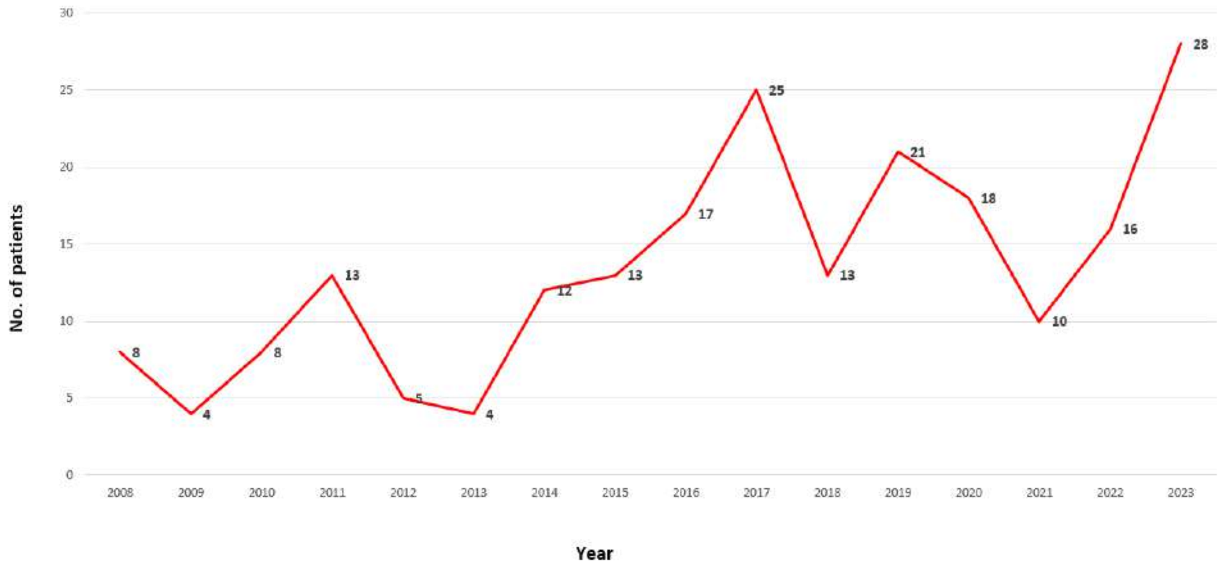

Figure 1: Trend of reporting of MDR/RR TB patients in Sri Lanka; 2008-2023

(NPTCCD, 2024)

The initial demographic pattern observed among MDR/RR TB patients are changing over time with marked differences in 2023. More cases are detected among the New TB cases with the scaling up of universal Drug Susceptibility testing. More females and younger age groups of twenties and early thirties are observed among the MDR/RR TB patients in 2023. Geographically more MDR/RR TB patients were initially reported from Western province and some districts such as Kandy and Ratnapura. However, by 2023, MDR/RR TB patients were reported from 12 districts.

The need to scale up universal Drug Susceptibility testing is a major requirement to bridge the gap between estimates and reported numbers.

1.3 New registration categories

According to the new classification by WHO, the patient registration categories are as follows.

1) New:

- A patient who has received no or less than one month of anti-TB treatment.

2) Non converter (Tx, Re Rx)

- A patient whose sputum was not converted at the end of the intensive phase.

3) Tx after loss to follow up (Tx, Re Tx):

- A patient who did not start treatment or whose treatment was interrupted for 2 consecutive months or more.

4) Treatment after failure (Tx, Re Tx):

-A patient whose treatment regimen had to be terminated or permanently changed to a new regimen or treatment strategy.

- *Reasons for the change include:*

- no clinical response and/or no bacteriological response;

- adverse drug reactions; or

- evidence of additional drug resistance to medicines in the regimen

bacteriological response refers to bacteriological conversion with no reversion:

- *"bacteriological conversion" describes a situation in a patient with bacteriologically confirmed TB where at least two consecutive cultures (for DR -TB and DS -TB) or smears (for DS-TB only) taken on different occasions at least 7 days apart are negative*
- *"bacteriological reversion" describes a situation where at least two consecutive cultures (for DR -TB and DS -TB) or smears (for DS-TB only) taken on different occasions at least 7 days apart, are positive either after the bacteriological conversion or in patients without bacteriological confirmation of TB.*

5) Relapse (Tx, ReTx):

- A patient whose most recent treatment outcome was “cured” or “treatment completed”, and who is subsequently diagnosed with bacteriologically positive TB by sputum smear microscopy, molecular tests or culture.

- Cured:

- A pulmonary TB patient with bacteriologically confirmed TB at the beginning of treatment who completed treatment as recommended by the national policy with evidence of bacteriological response and no evidence of failure.

- Tx completed:

- A patient who completed treatment as recommended by the national policy whose outcome does not meet the definition for cure or treatment failure

6) Tx after DRTB

- A patient who was previously treated for DRTB.

7) Other previously treated:

- This includes cases “transferred out” to another treatment unit and whose treatment outcome is unknown; however, it excludes those lost to follow-up

Chapter 2: DIAGNOSIS

Aim of this guidance is to update the TB case finding strategies and DR TB diagnostic algorithms as per the latest WHO recommendations and with the available diagnostic facilities with in the country.

At present, there are thirty-five (35) GeneXpert machines in the national TB diagnostic network. All the machines have been upgraded to perform Xpert Ultra and the testing is done exclusively with the GeneXpert Ultra cartridges.

GeneXpert testing is available in almost all the districts island wide. Three new 8 module GeneXpert machines with 10 colour (10 C) technology was added recently to the diagnostic network strengthening the capacity of testing.

NPTCCD has introduced the second WRD to the national laboratory network by utilizing the Truelab® machines by Molbio diagnostics which were imported to the country for Covid testing. These machines are currently placed at chest clinic laboratories.

Xpert MTB/XDR testing is available in the NTRL since 2023 and the capacity will be increased with availability of new machines with 10 C technology.

Currently at the NTRL, following methods are used for DST;

-Liquid media (automated BACTEC MGIT) for streptomycin(S), H, R, E and pyrazinamide(Z).

- Solid media (LJ medium) for first line drugs, isoniazid(H), rifampicin(R), and ethambutol(E) for certain indications

- Facilities are available for FL-LPA and SL-LPA. Genotypic DST for second line drugs including fluoroquinolone(FQ) is available with Xpert -XDR and SL-LPA.

At present, DST for Lzd and Bdq is not established and isolates are sent to WHO supranational laboratory (SRLN).

2.1 Currently available rapid molecular tests for TB diagnosis

1. Xpert MTB/RIF Ultra assay

The Xpert MTB/RIF Ultra assay (hereafter called Xpert Ultra) uses the same GeneXpert platform as the Xpert MTB/RIF test, and was developed to improve the sensitivity and reliability of detection of MTBC and RIF resistance. Xpert Ultra has a lower Level of Detection (LoD) ie.16 colony forming units [cfu]/mL) than Xpert MTB/RIF (131 cfu/mL).

At very low bacterial loads, Xpert Ultra can give a “trace” result, which is not based on amplification of the rpoB target and therefore does not give results for RIF susceptibility or resistance (may indicate as ‘indeterminate’). Therefore, additional tests such as phenotypic drug susceptibility testing is necessary to detect rifampicin susceptibility.

2. Truenat MTB, MTB Plus and MTB-RIF Dx assays

The Truenat MTB and MTB Plus assays use chip-based real-time micro-PCR for the semiquantitative detection of MTBC directly from sputum specimens and can report results in under an hour. The assays use automated, battery- operated devices to extract, amplify and detect specific genomic DNA loci.

If a positive result is obtained with the MTB or MTB Plus assay, an aliquot of extracted DNA needs to be tested with MTB-RIF Dx assays to detect mutations associated with RIF resistance. LoD of Truenat MTB, MTB Plus and MTB-RIF Dx tests are 100 CFU/ml, 29 CFU/ml and 200 cells/ml respectively. Truenat MTB, MTB Plus and MTB-RIF Dx assays will soon be introduced as the second WHO recommended rapid diagnostics (WRD) to the national tuberculosis laboratory network.

3. Xpert MTB/XDR

Xpert MTB/XDR detects MTBC DNA and genomic mutations associated with resistance to isoniazid, fluoroquinolones, ethionamide and second-line injectable drugs (amikacin, kanamycin and capreomycin) in a single cartridge.

Xpert MTB/XDR employs Cepheid’s GeneXpert platform, similar to that used by Xpert MTB/RIF and Xpert MTB/RIF Ultra. However, in the case of Xpert MTB/XDR, the platform supports multiplexing via 10-colour technology, which is different from the six-colour technology employed by Xpert MTB/RIF and Xpert MTB/RIF Ultra.

Xpert MTB/XDR should be performed on sputum samples which are known to be positive for MTB. The LoD of MTB by Xpert /MTB -XDR is 136 cfu /ml, higher than that of LoD of MTB Xpert Ultra (16 CFU/ml). Therefore, specimens with ‘MTB trace detected’ results when tested with the Xpert MTB/RIF Ultra assay are expected to be below the limit of detection of the MTB/XDR assay and are not recommended for testing with the Xpert MTB/XDR assay.

Xpert MTB/XDR can report results as “MTB not detected” or “MTB detected”. If results are reported as “MTB detected”, each drug is reported as resistance “detected” or “not detected”. If results are reported as “MTB not detected”, “invalid”, “error” or “no result”, then no DST results are reported.

Please note these molecular tests are not intended to monitor the response to treatment. Also, it is important to adhere to guidelines and algorithms when performing repeat testing with these molecular tests.

2.2 The current WHO recommendation

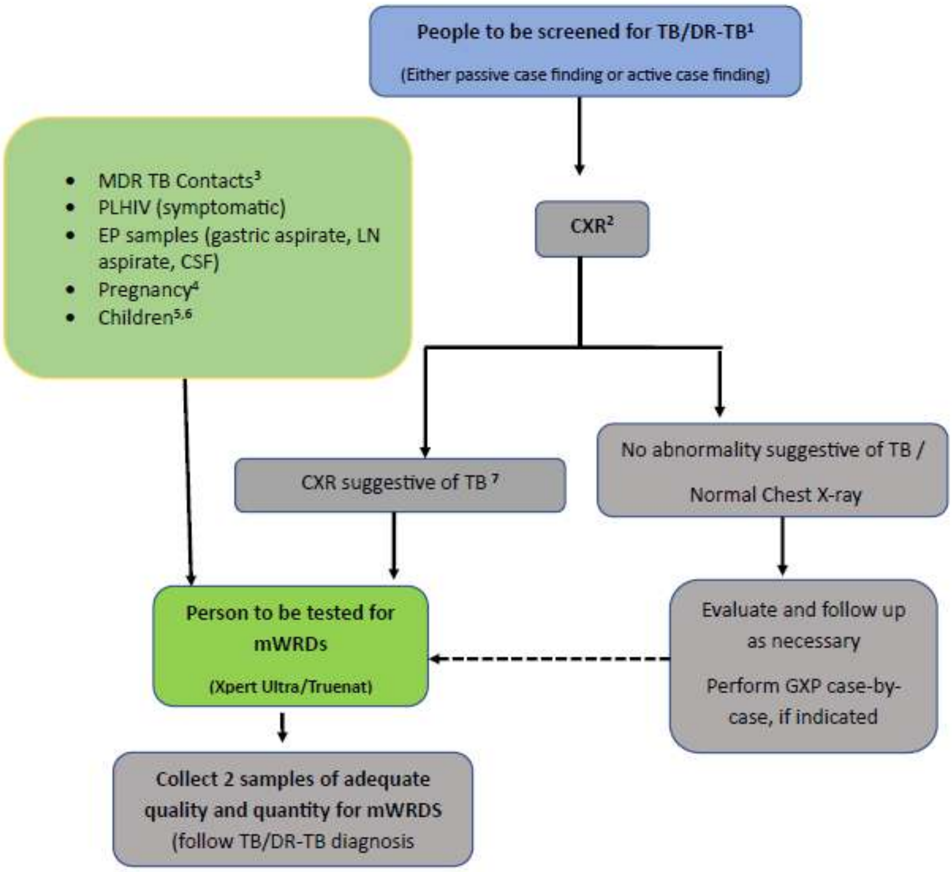

Figure 2: TB/DR-TB Case finding flow

Footnote

[1]People with any signs or symptoms of TB, systematic screening, contact screening and medical check-up

[2] If CXR is not available, samples to be tested using mWRDs (Xpert Ultra/Turenat) preferred if resources are available.

[3]all DR TB contacts and other high risk groups for TB are to be screened with CXR regardless of signs or symptoms. If CXR is not available, DR-TB contacts with TB signs or symptoms are to be tested for mWRDs

[4]CXR is contraindicated in the first trimester of pregnancy. Pregnancy with TB signs or symptoms, when CXR is not feasible, is to be tested for mWRDs regardless of sputum AFB result

[5]Presumptive pulmonary TB children <10 years with independent risk factors which are <2 years old, PLHIV, severe acute malnutrition is to be performed GXP

[6] Stool specimen/Gastric aspirates of presumptive TB children to be tested with Xpert Ultra

[7]CXR suggestive of TB includes ones with active or suspected, and healed lesions. In case of healed lesion, only patients with TB signs or symptoms are recommended to test with mWRDs.

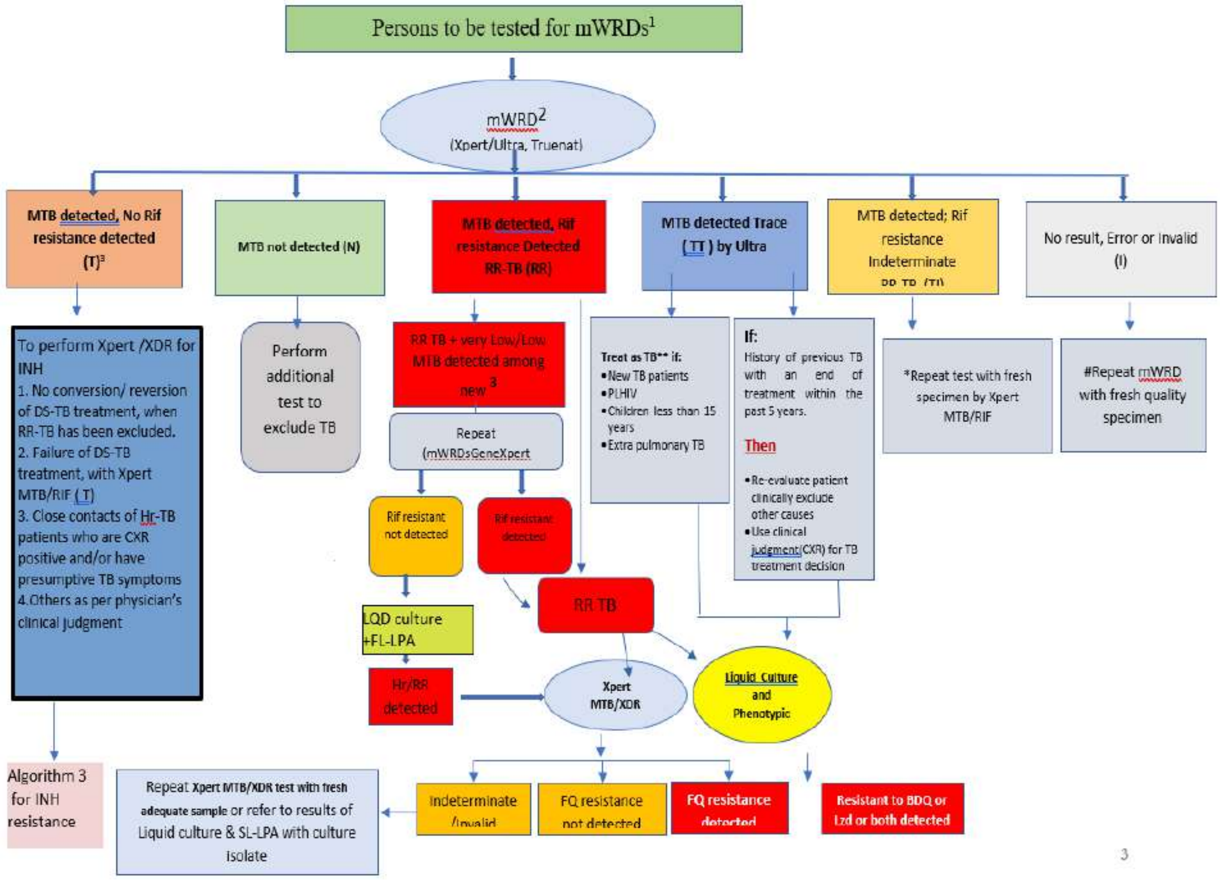

Figure 3: Algorithm 1 - TB/DR-TB Diagnosis Algorithm

Footnote:

[1]According to NTP's policy and criteria (see Figure 1.1: TB/DR-TB case finding flow)

[2]Molecular WHO recommended diagnostics (Xpert= Xpert MTB/RIF assay; Ultra=Xpert MTB/RIF Ultra assay; Others- Truenat or other centralized Diagnostic platforms). In future, only Xpert MTB/RIF Ultra and Truenat will be used for initial test. (To refer Genexpert testing criteria)

[3]Do not repeat the test in new patients with close contact to MDR TB, PLHIV, TB/DM, if MTB detected very low with RR detected.

[4]Liquid culture and phenotypic DST recommended

If Rif resistance (in patient with low risk for MDR/RR-TB), TI or I is detected in a repeat test, to proceed to first line LPA to get final Diagnosis.
* If Xpert test result rather than T (RR in low risk for MDR/RR-TB, N , TI , I) is found in repeat test , diagnosis will be confirmed by first line LPA
** Perform additional testing (CL/DST, LPAs) to rule out RR TB/FQ resistant, treat as MDR TB if close contact of MDRTB

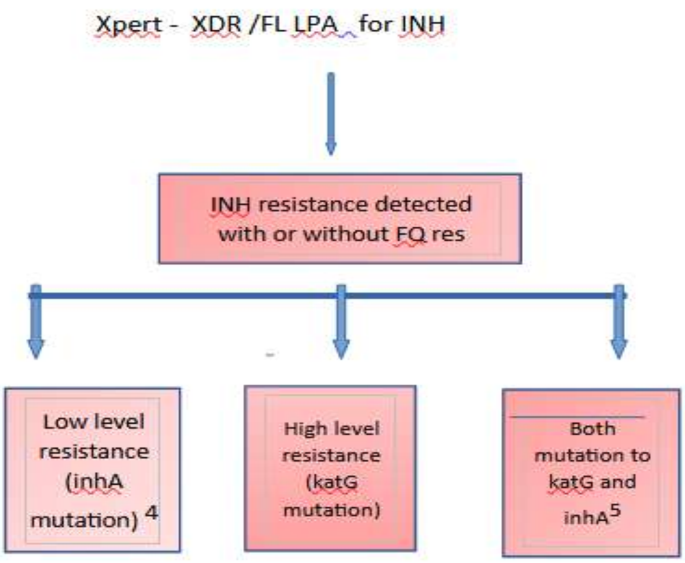

Figure 4: Algorithm 2 – INAH resistance

Foot notes:
[4] High dose INH is likely to be effective; Eto is unlikely to be effective.
[5] Not eligible to enroll in 9-month all-oral (9-Oral).

Interpretation of Discordant results

1. mWRD result "MTBC detected, RIF resistance detected"; RIF-susceptible by phenotypic DST:

a. Use the mWRD result to guide treatment decisions pending additional testing.

b. Borderline resistant mutations are known to generate this discordant result, particularly in the BACTEC mycobacterial growth indicator tube (MGIT) system (i.e. a false-susceptible phenotypic result). Patients infected with strains carrying these mutations often fail treatment with RIF-based first-line regimens.

In some settings with a low prevalence of MDR-TB, silent mutations have been observed that generate a false-resistant mWRD result, but these are rare.

c. False RIF-resistant results have been observed with the Xpert MTB/RIF cartridge when the MTBC detected result was "very low" and associated with probe binding delay. Follow-up action may include mWRD testing of the culture.

d. Follow-up actions may include DNA sequencing, confirmatory testing on other mWRD testing platform, phenotypic DST using solid media and evaluation of the possibility of laboratory or clerical error.

2. mWRD result "MTBC detected, RIF resistance not detected"; RIF resistant by phenotypic DST:

a. The treatment regimen should be modified based on the results of the phenotypic DST.

b. False RIF-susceptible mWRD results are rare but have been observed in 1–5% of RIF resistant TB cases tested with the Xpert MTB/RIF test in various epidemiologic settings. Mutations in the region of the rpoB gene sampled by the Xpert MTB tests have been shown to account for 95–99% of RIF resistance. The remainder of RIF resistance arises from mutations outside the sampled region, which produce an Xpert MTB result of "RIF resistance not detected".

Chapter 3: INTRODUCTION TO NEWER REGIMEN IN DRTB – BPALM/ BPAL

Since our last updated guideline in 2021 some more evidence has emerged regarding the treatment of drug-resistant TB through the recent Ze Nix trial, Nix TB trial and TB Practical trial.

These studies have shown that shorter regimens containing Bdq, Pa, Lzd and Mfx given for six months can lead to successful bacteriological clearance and disease-free survival extending beyond 24 months.

Concerns were raised about the long-term use of Lzd as it is known to cause neurological and hematological side effects. Zenix study has shown that 26 weeks (6 months) of Lzd 600mg /day is a safe option.

3.1. 2022 DR TB key recommendations

1. **6 months BPaLM** for people 14 years or above with MDRTB having susceptibility to fluoroquinolones or when FQ resistance is unknown.

(Still there is no clear evidence supportive of replacing Mfx with Lfx in this regimen)

2. **6 to 9 months of BPaL** for patients 14 years or above years having MDRTB with FQ resistance
3. **9 to 12 months of oral shorter regimen)** either Ethionamide(Eto) or Lzd based regimen if the eligibility criteria are fulfilled (see page 33 national guideline 2022)

   Bdq-6, Eto-6, Hh -6, E-11, Z-11, Cfs-11, Lfx-11 or

   Bdq-6, Hh-6, Lzd-2 E-11 Z-11 Cfs-11 Lfx-

   Lzd based regimen is recommended for pregnant or lactating mothers.

4. **18 months individualized longer regimen** is used when none of the above regimens could be used

In addition to the above, certain other regimen are recommended by WHO in 2025 guidelines for Treatment & Care, which could be used by the clinicians when deciding on a treatment regimen (Annexure I). **The 6-month bedaquiline, delamanid, linezolid, levofloxacin and clofazimine (BDLLfxC) regimen and 9-month all-oral shorter regimen are safe to use in children as Pa is not included in them.**

3.2 Actions before initiating treatment

1.) Good history

- past episodes of TB & treatment received
- co-morbidities & their treatment
- drug allergies
- Smoking
- alcohol consumption
- family background/support, employment, socio economic status

2.) Examination

i) General examination (features of organ failure etc.)
ii) Peripheral neuropathy
iii) Visual impairment
iv) Psychological assessment

3.) Baseline investigations in BPaLM

i) CXR (other radiological investigations where appropriate)
ii) FBC
iii) FBS
iv) S electrolytes
v) S bilirubin
vi) SGPT
vii) Serum Creatinine
viii) ECG
ix) echocardiography (where appropriate),
x) HIV status
xi) pregnancy test (where appropriate),
xii) Thyroid function tests

3.3 Inclusion/Exclusion Criteria for BPaLM/BPaL

a) Inclusion criteria

- Adults and adolescents aged 14 years and older regardless of HIV status
- Patients with confirmed MDR/RR-TB (BPaLM) or with MDR/RR-TB and confirmed resistance to fluoroquinolones (pre-XDR-TB for BPaL)

- No known allergy to any of the drugs in BPaLM regimen
- Pulmonary TB and all forms of extrapulmonary TB except for TB involving CNS, osteoarticular and disseminated (miliary) TB
- Patients with less than 1-month previous exposure to Bdq, Lzd, pa or Delamanid(Dlm). When exposure is greater than 1 month, these patients may still receive these regimens if DST to these medicines shows susceptibility

b) Exclusion Criteria

- Under 14 years of age
- Confirmed XDR TB patients (MDR/RR-TB that is also resistant to any of FQs, and either Bdq or Lzd or both)
- Known resistance to either Bdq, Pa, or Lzd, Dlm or previous exposure of these drugs more than one month in the absence of susceptible DST results
- Pregnant and breastfeeding women as limited safety data is available for use of Pa in this group of patients
- If patient has one or more of the following medical conditions:

- Baseline QTcF of > 500 ms which cannot be corrected
- Baseline Grade 3-4 peripheral neuropathy
- Baseline severe anaemia (Hb < 8.0 g/dl which is un-correctable)
- Baseline moderate thrombocytopenia (Platelet <75000/ mm3)
- Baseline moderate neutropenia (Neutrophil count <1000/mm3)
- Baseline ALT/AST > 3 x upper limit of normal

c) Caution

People Living with HIV -due to possible drug interaction with antiretroviral drugs (page 43 national guideline 2022)

Pregnancy and breast feeding -due to limited evidence

3.4 Mechanism of Actions of Drugs used in BPaLM/BPaL against drug-resistant *Mycobacterium tuberculosis*

BPaLM/BPaL regimen are highly bactericidal and makes it very effective against drug resistant TB bacteria as follows.

- **Bdq** is a diarylquinoline that blocks adenosine triphosphate (ATP) synthase,
- **Pa** is a nitroimidazole that inhibits cell wall biosynthesis,
- **Lzd** is an oxazolidinone that inhibits protein synthesis,
- **Mfx** is a fluoroquinolone that inhibits the mycobacterial topoisomerases.

Table 1: Drug regimen in BPaLM

Drug	Dose	Number of Tablets (26 weeks)
Bedaquiline (100 mg tablets)	400 mg once daily for 2 weeks, followed by 200 mg 3 times per week for 24 weeks	200
Pretomanid (200 mg tablets)	200 mg once daily	182
Linezolid (600 mg tablets)	600 mg once daily (adjustable only after 9 weeks)	182
Moxifloxacin (400 mg Tablet)	400 mg once daily in case of BPaLM	182

3.5 Treatment Duration of BPaLM/BPaL

- BPaLM is recommended for 6 months only (26) weeks,
- BPaL is recommended for 6-9 months (26-39 weeks) months and can be extended up to 9 months because of the following.
  - sputum cultures are positive between months 4 and 6 of treatment or
  - Delayed radiological response by 6 months unless there is other indication to stop the treatment
- If regimen is shifted from BPaLM to BPaL, then treatment duration can be extended up to 39 weeks and treatment initiation date will be counted as started date of BPaLM

3.6 Management after Treatment interruptions of BPaLM

i) Any treatment interruption of the whole regimen longer than 7 days to one month can be added at the end of treatment

ii) Management of Bdq missed Doses in maintenance phase

   a. Missed dose < 14 days: Bdq should be administered as soon as possible, no Bdq reloading is required.

   b. Missed dose > 14 days: If BPaLM/BPaL is interrupted for more than 2 consecutive weeks, shifting to individualized longer regimen/ 9-month oral regimen will be considered

iii) Rescheduling of Bdq: If Bdq is dosed every Monday, Wednesday and Friday, then if the Wednesday dose is missed it can still be taken on Thursday, and then the following dose should be taken on Saturday, with a return to the usual dosing schedule on Monday.

Chapter 4: MODIFICATIONS IN BPaLM/BPaL AND MANAGEMENT OF ADVERSE EVENT/ ADVERSE DRUG REACTION

4.1 Modifications in BPaLM/BPaL

In case of severe/serious toxicity consider the following for modifications;

- The modifications in dosages or stopping of Bdq/ Pa only is not allowed at any point of treatment
- Lzd 600 mg dosages can be modified/dose reduced only after 9 weeks of treatment
- If there is Lzd toxicity during 1st 9 weeks of treatment, the whole regimen is to be temporarily withheld.
- While after 9 weeks of treatment if there is Lzd toxicity, only Lzd can be temporarily stopped, or dose reduced
- If Lzd has to be stopped in the last 8 weeks of treatment, complete the course with the remaining component drugs- For example after 18 weeks or 4 months of treatment if Lzd is permanently stopped, rest of the regimen to be continued

4.2 Reintroduction of regimen after temporary interruption

When the whole regimen is temporarily stopped for suspected drug toxicity at any point of treatment, it can be reintroduced if;

- There is no more than 14 days of consecutive treatment interruption or
- Up to a cumulative 4 weeks of non- consecutive treatment interruption
- Moxifloxacin alone is discontinued at any point because of toxicity, the regimen can be continued as the BPaL regimen.

4.3 Permanent cessation of the whole regimen

- If Linezolid has to be permanently discontinued during the initial 18 weeks of treatment
- If any one of Bdq or Pa (at any point of treatment) is permanently discontinued

4.4 Managing Lzd Toxicity

Linezolid (Lzd) causes the most and frequent toxicities and is responsible for AEs up to 50%. The most common toxicities related to Lzd are myelosuppression, peripheral neuropathy and optic neuritis. In Nix TB Trial which used 1200 mg of Lzd, 47% of patients experienced myelosuppression and about 80% of patients suffered from peripheral neuropathy and 11% reported Optic neuritis mostly with grade 1 and 2 which resolved. Myelosuppression is usually Lzd dose related and may occur in early weeks of treatment and Peripheral neuropathy is both dose and duration related and usually occurs after 8 weeks of treatment. Lzd toxicity is easy to detect and manage but requires close monitoring under a well-functioning aDSM mechanism.

4.4.1 Optic Neuritis

Optic neuritis is relatively serious adverse effect of Lzd but not so common. This is not easy to detect especially if the patient does not complain of visual problems. Also, this can present as an asymptomatic occurrence with only a change in the visual acuity testing. If patient during treatment complains of blurred vision or eye pain, such patients should be further assessed and referred to ophthalmologist. In Myanmar, only 1 out of 100 patients from BPaL OR cohort reported optic neuritis (Adverse Event monitoring up to July 2023).

Symptoms of Optic Neuritis: Optic neuritis may occur unilateral or bilateral and symptoms may appear suddenly or gradually. More severe the nerve damage, the more serious the symptoms. If Lzd is stopped early once symptoms appear then it is reversible, if there is delay in stopping the drug then the chance of permanent optic nerve damage is high.

Following are main symptoms of optic neuritis (American academy of Ophthalmology)

- Diminished vision (usually the main symptom)
- Trouble distinguishing colours(specially loss of red-green colour distinction), or noticing that colours aren't as vibrant as usual
- Vision that appears blurry
- Inability to see out of one eye
- Abnormal reaction of the pupil when exposed to bright light
- Pain in the eye, especially on eye movement

Management

Optic neuritis is best tested using the Ishihara test or a visual acuity test (e.g., Snellen chart) should be used to detect any sign of visual impairment. Optic neuritis diagnosed at any grade, permanently discontinue Lzd and always refer to an ophthalmologist for opinion. Lzd should only be restarted if advised by ophthalmologist. Diabetics with uncontrolled glycaemia are at high risk of neuritis, so aim for proper glycaemic control during MDR TB treatment.

Important Practice Points:

Optic neuritis will rarely happen during the first 2 months of treatment.

If optic neuritis is suspicious (by symptoms and/or Vision test), the following should be considered for further management;

1. During the first 9 week of treatment, withhold BPalM/BPal regimen and refer patient to ophthalmologist

- ➢ If optic neuritis is not diagnosed, BpalM/BPal regimen can be reintroduced
- ➢ If optic neuritis is confirmed, shift patient to individualized longer regimen

2. After 9 weeks of BPaLM/BPaL treatment, only Lzd can be temporarily interrupted and refer to ophthalmologist

- ➢ If optic neuritis is not diagnosed, reintroduce Linezolid and continue the treatment
- ➢ If optic neuritis is confirmed,
    - in treatment period less than 18 weeks – discontinue BpalM/Bpal and switch to individualized oral longer treatment regimen
    - in treatment period more than 18 weeks of treatment – omit Lzd and can continue treatment with remaining drugs

Note: If treatment interruption of BPalM/BPal regimen is more than two consecutive weeks, shifting to individualized longer regimen will be considered

4.4.2 Peripheral Neuropathy (PN)

Peripheral neuropathy is a serious condition characterized by symmetrical, distal damage to the peripheral nerves. Neurotoxicity is one of the major factors leading to the temporary interruption or permanent withdrawal of linezolid for some patients. Numerous studies have described the relief of neuropathic symptoms after linezolid has been discontinued (Dempsey et al 2018). The type of neuropathy caused by Lzd or chemotherapy is labelled as toxic neuropathy with increased sensitization of nerves. TB/DR TB itself or HIV infection, use of alcohol can cause peripheral neuropathy where some patients may present symptoms of burning, tingling or numbness at baseline. Moreover, pre-existing DM can also cause Diabetic neuropathy and all these factors can further precipitate Lzd related toxicity. However, with Lzd related toxicity in addition to tingling, numbness patients may present symptoms of pins and needles sensation or electric or stabbing pain in lower or upper extremities. Grade 1 and 2 PN at baseline is not contraindication for BPaLM/BPaL initiation.

In the Nix TB study 81% of patients experienced some level of Peripheral neuropathy, while in Myanmar BPaL study 33% of patients reported various degrees of peripheral neuropathy. It has also been observed from Nix TB study and from Myanmar BPaL study that 600 mg daily dose of Lzd is relatively safe in programmatic conditions. Lzd toxicity is generally dose and duration related so, mostly its neurotoxic effect starts after or by 8 weeks of treatment initiation.

The following are required for assessment of PN;

Physical examination is performed by a health professional which includes;

a) Listening to the patient for signs/symptoms of peripheral neuropathy (pain, aching, burning, tingling or numbness, pins and needles sensation or electric or stabbing/shooting pain in upper or lower extremities, may or may not be exacerbated from stimuli). Observe any changes in presenting complaints from baseline or from previous month.

b) assessment of the patient's feet on appearance and ulceration

c) ankle reflexes,

d) light touch perception with a cotton swab / monofilament

e) vibration perception using a 128 Hz tuning fork. A patient is considered peripheral neuropathic when their score is ≥2 (on an 8-point scale).

Please remember that this assessment is subjective and may lead to wrong scoring if it is not properly carried out. Therefore, it is imperative that doctors/nurses/health care workers should be appropriately trained to carry out below assessments. For assessing PN grading procedures, please refer to Annex 1.

If Peripheral Neuropathy Grade 1-2:

Grade 1= sometimes asymptomatic or mild *discomfort with BPNS subjective sensory scoring from 1-3 on any side on examination/assessment*

Grade 2= *moderate discomfort with BPNS subjective sensory scoring from 4-6 on any side*

- If it occurs during the first 9 weeks of treatment, only Lzd interruption or dose reduction is not allowed, hence, the whole BPaLM/BPaL regimen to be stopped for 7-14 days (one to 2 weeks). Reassess after 7-10 days and if symptoms improve, then re-introduce whole regimen with 600 mg of Lzd, monitor closely, if symptoms worsen then stop whole regimen and shift to OLTR or 9-month all-oral regimen (Eto variation) if the BPaLM/BPaL treatment duration is less than one month. Though, PN during the first 2 months is not common.

- If it occurs after 9 weeks of BPaLM/BPaL treatment then, temporarily stop/interrupt Lzd with drug holiday for 7-14days (one to 2 weeks) and then reduce dose to 300 mg daily, monitor closely after dose reduction and if symptoms worsen (Grade 3 or 4), then stop Lzd 300 mg. If condition reverses back to grade 1 or 2, then reintroduce 300 mg and monitor. If symptoms worsen, permanently stop Lzd.

- If Lzd is permanently stopped, assess the treatment period -
    - in treatment period less than 18 weeks – discontinue BpalM/Bpal and switch to individualized oral longer treatment regimen
    - in treatment period more than 18 weeks of treatment – omit Lzd and can continue treatment with remaining drugs

If Peripheral Neuropathy Grade 3-4:

Grade 3= *severe paraesthesia, severe discomfort, intolerance with BPNS subjective sensory scoring from 7-10 on any side*

Grade 4= *incapacitating, disabling not able to perform daily routine activities*

- Stop Lzd, most patients require long term suspension of Lzd
- If it occurs during the first 9 weeks of treatment, only Lzd interruption or dose reduction is not allowed, hence, the whole BPaLM/BPaL regimen to be stopped. Reversal to grade 1 or 2 is time taking process and may not improve in 7-14 days, in most cases, patient has to be shifted to OLTR
- If it occurs after 9 weeks of BPaLM/BPaL treatment then, temporarily stop/interrupt Lzd with drug holiday for 7-14days (one to 2 weeks) and then reduce dose to 300 mg daily if symptoms improve and reverse back to grade 1 or 2. Monitor closely after dose reduction and if symptoms worsen back to grade 3-4, then stop Lzd 300 mg.
- If Lzd is permanently stopped, assess the treatment period -
    - in treatment period less than 18 weeks – discontinue BpalM/Bpal and switch to individualized oral longer treatment regimen
    - in treatment period more than 18 weeks of treatment – omit Lzd and can continue treatment with remaining drugs

Peripheral Neuropathy Treatment for symptomatic relief

- In grade 1- Increase the dose of Pyridoxine to 200 mg daily and prescribe Paracetamol/acetaminophen along with counselling
- In grade 2-3- prescribe NSAIDs as they are helpful in reducing the inflammatory sensitization of nerves, if required prescribe Gabapentin (100-300 mg every night or 100-300 mg thrice daily) or Pregabalin 75 -150 mg in BD doses
- Consider trying Tramadol if required especially when there is severe intolerance to pain and other measures not working, but in most cases drug cessation leads to gradual symptoms relief.
- Counsel the patients and explain that the causative agent has been stopped and we will review after some time, and it is in most cases reversible after some time.
- For Diabetics, aim to control hyperglycaemia for infectious patients, and Insulin is the choice of treatment

4.4.3 Myelosuppression

Myelosuppression is one of the toxic effects of Lzd causing suppression of bone marrow resulting in decreased red blood cells, white blood cells and platelet counts. In Myanmar BPaL study reported that 48% of patients experienced myelosuppression of various degree but often mild to moderate (grade 1 and 2) and similarly, Nix TB study reported 47% myelosuppression. However, Lzd with 600 mg showed lesser degree of myelosuppression as compared to 1200 mg daily dosage as reported by ZeNix, TB PRACTECAL and Myanmar BPaL study.

Lzd related myelosuppression is generally dose related and may occur as early as within 2 weeks or mostly within 3 months. However, Lzd associated myelosuppression is completely reversible on its temporary cessation/withdrawal.

At baseline some patients may present with anaemia mainly due to TB/DR-TB infection, HIV, malabsorption, iron/folate deficiency, occult GI bleeding. Grade 1 and 2 anaemia is not a contraindication for BPaLM/BPaL. Effective DR TB therapy may improve baseline anaemia status.

If there is a decrease in Hb% level of 10% or more after 4 weeks of treatment, then such patients are at high risk for severe aneamia.

Table 2: Step wise approach for management of patients with Myelosuppression

Severity Grade	**Grade 1** **Mild**	**Grade 2** **Moderate**	**Grade 3** **Severe**	**Grade 4** **Life-threatening**
Anaemia	10.5 - 9.5 g/dL	9.4 – 8.0 g/dL	7.9 – 6.5 g/dL	< 6.5 g/dL
Platelets decreased	99,999-75,000/mm³	74,999-50,000/mm³	49,999-20,000/mm³	< 20,000/mm³
While blood cells decreased	<LLN - 3,000/mm³	<3,000 - 2,000/mm³	<2,000 - 1,000/mm³	< 1,000 /mm³
Absolute neutrophil count low	1500 - 1000/mm³	999 - 750/mm³	749 - 500/mm³	<500/mm³
Action	Monitor carefully and consider reduction of dose of Lzd (300 mg daily except in BPaLM/BPaL during 1st 9 weeks of treatment). Drug holiday for 7-14 days before dose reduction may help in reducing myelosuppression	Monitor carefully, perform CBC weekly and consider reduction of dose of Lzd (300 mg daily or 600 mg thrice weekly), in case of Grade 2 neutropenia, stop Lzd immediately. Restart at reduced dose once toxicity has decreased to Grade 1. In case of Grade 2 anemia, consider erythropoietin	Stop Lzd immediately. In case of Grade 3 anemia, consider EPO if available. Restart at reduced dose once toxicity has decreased to Grade 1 or consider stopping Lzd permanently	Stop Lzd immediately. Hospitalize patient and consider blood transfusion or EPO. Restart at reduced dose once toxicity has decreased to Grade 1 or consider stopping Lzd permanently

		(EPO). Drug holiday for 7-14 days before dose reduction may help in reducing myelosuppression		

If Myelosuppression Grade 1-2:

- If it occurs during 1st 9 weeks of treatment, then only Lzd interruption or dose reduction is not allowed, whole BPaLM/BPaL regimen to be continued (except in grade 2 neutropenia) with 600 mg daily. Monitor closely and if myelosuppression worsens to grade 3 or 4, stop BPaLM/BPaL.
- If it occurs after 9 weeks of BPaLM/BPaL treatment, continue 600 mg of Lzd, monitor closely and consider dose reduction to 300 mg daily where risk of further myelosuppression is high. Despite with reduced dosage if toxicity worsens to grade 3-4 then stop Lzd and do not restart and patient is shifted to OLTR.
- If Lzd is permanently stopped, assess the treatment period -
  - in treatment period less than 18 weeks – discontinue BPalM/BPal and switch to individualized oral longer treatment regimen
  - in treatment period more than 18 weeks of treatment – omit Lzd and can continue treatment with remaining drugs

If Myelosuppression Grade 3-4:

- If it occurs during 1st 9 weeks of treatment then only Lzd interruption or dose reduction is not allowed, whole BPaLM/BPaL regimen to be stopped for 7-14 days (one to 2 weeks). Some patients may require blood transfusion. Reversal from grade 4 to normal may take considerable time and assess patients' case by case to decide whether to stop BPaLM/BPaL permanently. Reassess CBC after 10 days and if result shows grade 1-2 myelosuppression (except grade 2 neutropenia) then re-introduce whole regimen with 600 mg of Lzd, monitor closely, if it worsens to grade 3-4 again, then stop whole regimen and shift to OLTR. During 30 days of treatment patient can be shifted to Eto based OSSTR if eligible. There is possibility that grade 3-4 myelosuppression may happen during early weeks of treatment

- If it occurs after 9 weeks of BPaLM/BPaL treatment then, temporarily stop/interrupt Lzd with drug holiday for 7-14days (one to 2 weeks). If myelosuppression improves to grade 1-2, then reduce dose to 300 mg daily, monitor closely after dose reduction and if CBC shows worsening to grade 3-4, then stop Lzd 300 mg. If condition reverses back to grade 1-2 then reintroduce 300 mg and monitor, if toxicity continues with 300 mg, then do not restart and patient is shifted to OLTR.

- If Lzd is permanently stopped, assess the treatment period -

    - Treatment period less than 18 weeks – discontinue BPalM/BPal and switch to individualized oral longer treatment regimen

    - Treatment period more than 18 weeks of treatment – omit Lzd and can continue treatment with remaining drugs

Important points to consider

- Investigate causes of baseline myelosuppression and treat accordingly, optimize management of co-morbidities.

- Evidence of iron deficiency anaemia at baseline is having low ferritin, microcytosis, hypochromia, if needed perform serum electrophoresis.

- Give oral iron supplements, ferrous sulphate 200 mg BD, if not tolerated ferrous fumarate to be used. Prescribe ascorbic acid 250-500 mg BD for better iron absorption.

4.5 Management of QT Interval Prolongation

Moxifloxacin (Mfx) to a greater degree and Bdq to a lower degree are the most causative agents for cardio toxicity in BPaLM /BPaL regimens. While other causes could be electrolyte imbalance and hypothyroidism. In Nix TB study and Myanmar BPaL study reported 6% patients suffered from QTc interval prolongation but none of them had to discontinue BPaL.

Spontaneous visits should be arranged for those patients experiencing any clinical symptoms suggestive of cardiotoxicity such as tachycardia, syncope, palpitations, weakness or dizziness and ECG to be performed. If QTc is prolonged, assess for arrhythmias and send investigation for electrolyte checking. If needed, hospitalize the patient to monitor.

Following is step wise approach for prolonged QTc interval management.

Normal values of QTcF: Male (M): <450 ms and Female (F): <470 ms

If QTc interval prolongation is Grade 1-2:

QTcF Grade 1-2= 450-500 ms:

- If asymptomatic, continue treatment, monitor more closely; at least weekly ECG until QTcF has returned to less than grade 1/normal. Replete electrolytes as necessary. Rule out other causes.

If QTc interval prolongation is Grade 3-4:

QTcF Grade 3 = QTcF more than 500 ms on repeat ECG at least after 30 minutes, without signs / symptoms of arrhythmia

QTcF Grade 4 = QTcF more than 500 ms or >60 ms change from baseline and one of the following: (Torsade de pointes or polymorphic ventricular tachycardia or signs / symptoms of serious arrhythmia)

- Stop the BPaLM/BPaL regimen and all other suspected causative drugs, Hospitalize, check, and replace electrolytes as necessary, including non-TB drugs, check for other potential causes and manage accordingly. Repeat ECG after ≥24 hours but < 48 hours, until QTcF < 500 ms

Important Practice Points:

- If BPaLM/BPaL is stopped due to QTcF prolongation, then it can be reintroduced provided that
  a. If whole regimen did not stop more than 14 consecutive days during 1st 9 weeks of treatment initiation
  b. If whole regimen did not stop more than 4 weeks cumulatively during course of treatment
- Reintroduction of Mfx and Bdq should be cautious. If patient is on BPaLM and it necessitates to stop Mfx permanently due to toxicity, switch patient to BPaL
- If patient is on BPaL and it necessitates to stop Bdq permanently at any point of time, then switch patient to OLTR
- Only Bdq interruption is not allowed at any time during treatment

4.6 Management of Hepatotoxicity

Hepatotoxicity in patients treated with the BPaLM/BPaL regimens can be related to Bdq and Pa. In Myanmar BPaL study, 51% of patients experienced transaminase increase and were mild to moderate in nature.

Table 3: Managing elevated liver enzymes according to severity grading

Grade **Severity**	**Grade 1** **Mild**	**Grade 2** **Moderate**	**Grade3** **Severe**	**Grade 4** **Life-threatening**
ALT /AST	>ULN – 3.0 x ULN	>3.0 – 5.0 x ULN	>5.0 – 20.0 x ULN	>20.0 x ULN
Bilirubin	>ULN - 1.5 x ULN	>1.5 - 3.0 x ULN	>3.0 - 10.0 x ULN	>10.0 x ULN

Action	Continue treatment regimen. Patients should be followed until resolution (return to baseline) or stabilization of AST/ALT elevation	Continue treatment regimen. Patients should be followed until resolution (return to baseline) or stabilization of AST/ALT elevation	Stop full BPaLM/BPaL regimen, including other non-TB drugs; measure LFTs weekly. Treatment may be reintroduced after toxicity is resolved, (liver enzymes returned to Grade 1)	Stop full BPaLM/BPaL regimen, including other non-TB drugs; measure LFTs weekly. Treatment may be reintroduced after toxicity is resolved, (liver enzymes returned to Grade 1)

Important Practice Points:

Due to grade 3 and 4 liver toxicity if BPaLM/BPaL is stopped, then it can be reintroduced if;

- It did not stop more than 14 consecutive days during 1st 9 weeks of treatment initiation
- It did not stop more than 4 weeks cumulatively during course of treatment
- If unable to restart with in the above timelines, then shift patient to OLTR
- Only Bdq or Pa interruption is not allowed

4.7 Managing other adverse events

All major and most common adverse events have been covered in above sections. All other relevant drug adverse events should be managed as per DR TB guidelines, severity grading as per aDSM chapter.

4.8 Shifting Patients from BPaLM/BPaL to OLTR/OSSTR

In following cases switch patient to OLTR/OSSTR;

- Consecutive treatment interruption of full regimen more than 2 weeks
- A cumulative of non-consecutive treatment interruption more than 4 weeks of whole regimen
- If only Pa or Bdq is to be permanently stopped at any point of time

- During 30 days of treatment patient can be shifted to Eto based OSSTR if eligible.
- If treatment period less than 18 weeks and Lzd toxicity is not manageable – discontinue BpalM/Bpal and switch to individualized oral longer treatment regimen

Important Practice points:

- Once switching to OSSTR/ OLTR due to drug toxicity, patient is to be declared failure
- Review the patient clinically
- If treatment response is adequate, the existing drugs to be considered likely effective while designing OLTR.

The Side Effect Monitoring and Management during DR-TB Treatment format (DRTB 6) should be maintained in every DRTB patient file

Any Side Effect reported should be reported to national level through SE format and Severe Adverse Event (SAE) format accordingly.

Chapter 5: MONITORING OF BPaLM/BPaL PATIENTS

All patients treated with the BPaLM/BPaL regimens should be assessed properly and to be evaluated at baseline, during and after treatment, including clinical evaluation, bacteriological and laboratory testing, according to the monitoring schedule (Table 4.1 & 4.2 below).

Following is stressed:

1. Monthly assessment appropriate management of co-morbid conditions specially HIV and Diabetes. Insulin to be prioritised for DM patients during acute infectious state.
2. All possible measures should be taken to provide Patient entered follow up and DOT care as otherwise it may result in loss to follow up.
3. Patient centred approach to be adopted along with monthly social and psychological support with proper mental health care

Table 4: Monitoring schedules for BPaLM/BPaL regimens

Examination	Baseline	2 Weeks	Monthly	End of Treatment	6- and 12- months post-treatment
Clinical Evaluation					
Clinical Assessment	X	X	X	X	X
Weight and BMI	X	X	X	X	X
Peripheral neuropathy	X	X	X	X	
Visual acuity and Ishihara test	X	X	X	X	
Assessment and follow up of Adverse Events	X	X	X	X	X
Outcome evaluation				X	X
Radiology, ECG, and other Lab Tests					
Chest X-Ray	X			X	X
ECG	X	X	X	X	
Full Blood Count (FBC)	X	X	X	X	
Liver Function Tests	X		X	X	
Serum electrolytes	X		X	X	

Urea, Creatinine	X				
Serum Amylase *					
S. Lactate *					
Pregnancy Test	X				
HIV /HBV/ HCV	X				
Fasting Blood	X				

*As indicated

NOTE: More frequent monitoring may be advisable in specific situations, including elderly people, patients infected with HIV, affected by HBV- or HCV-related hepatitis, diabetes mellitus, or with moderate to severe hepatic or renal impairment

Table 5: Monitoring BPaLM/BPaL patients by smear and culture

Tests	**Month**									
	0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
Smear	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Liquid culture	✓	✓	✓	✓	✓	✓	✓	✓*	✓*	✓*

NOTE: * culture of 7 ,8,9 will be applied if treatment is extended beyond 6 months

Post Treatment Outcome Follow up;

Patients should also be advised to return for post-treatment follow up 6 and 12 months after completion of treatment, which includes a clinical evaluation, CXR and if needed Culture and DST. In a patient who has a positive culture documented during the post-treatment follow-up period, a post-treatment DST should be performed, including for Bdq, Pa and Lzd

Chapter 6: DECLARING TREATMENT OUTCOME IN BPaLM/BPaL

Table 6: Treatment outcomes in BPaLM BPaL

Treatment failed	A patient whose treatment regimen needed to be terminated or permanently changed [(a)] to a new regimen or treatment strategy.
Cured	A pulmonary DR TB patient with bacteriologically confirmed TB at the beginning of treatment who completed treatment as recommended by the national policy (6 month for BPaLM and 6- 9 months for BPaL), with 2 consecutive negative culture during last 3 months of treatment (4,5 or 5,6) and no other circumstantial features suggesting possibility of failure such as clinical deterioration, radiological deterioration despite the bacteriological response.
Treatment completed	A patient who completed treatment as recommended by the national policy, whose outcome does not meet the above definition for cure or treatment failure
Lost to follow-up	A patient who did not start treatment or whose treatment was interrupted for 2 consecutive months or more.
Died	A patient who died for any reason before starting treatment or during the course of treatment
Not evaluated	A patient for whom no treatment outcome was assigned- This includes cases "transferred out" to another treatment unit and those whose treatment outcome is unknown; however, it excludes those lost to follow-up
Treatment success	The sum of cured and treatment completed

[(a)]Reasons for the change include:

- ***no bacteriological response [(*)]***
- ***Adverse drug reactions; or***
- ***Evidence of additional drug resistance to medicines in the regimen.***

[(*)] Persistent culture positive (no conversion) up to 4 months cultures or bacteriological reversion after conversion

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Chapter 7: CONTACT SCREENING & PROPHYLACTIC TREATMENT

The contact screening of DRTB patients has to be carried out same as per DSTB patients. The initial screening tool is CXR. For sputum examination, use of NAAT is compulsory as this is a high-risk group for MDR TB. For more details refer National Manual for TB Control, 2021 update (Chapter 3). For those negative for TB Disease, should be tested for TB infection as per the Latent TB management guidelines, NPTCCD 2021.

Based on the type of resistance of the index case, prophylactic treatment of close contacts should be decided.

Table 7: Recommended TPT regimen based on type of resistance

Type of resistance in Index case	Suggested TPT	Description
1.) H resistance with R sensitivity	4 R	If the index case has H resistance with investigated R sensitivity, R for 4 months can be used.
2.) For contacts of R resistance, INH sensitive – 6H	6H	If the index case has R resistance with investigated H sensitivity, H for 6 months can be used.
3.) MDR	6Lfx	Levofloxacin(Lfx) 6m after excluding Lfx resistance among the index case can be used.
4.) Pre Xdr & MDR	No prophylaxis suggested	

(WHO, 2024)

According to WHO PMTPT guidelines 2024 (WHO consolidated guidelines on tuberculosis, 2024; WHO operational handbook on tuberculosis, 2024), Lfx for 6 months is recommended for contacts of MDR TB patients *(Strong recommendation, moderate certainty of the estimates of effect)*. This is based on two randomized controlled trials, TB CHAMP and V-QUIN, and a systematic review commissioned by WHO on TPT for MDR/RR-TB (Annex 5). In addition, studies on the programmatic feasibility and acceptability of 6Lfx were conducted.

Subgroup considerations

Children and adolescents:

Lfx appears to be more tolerable among children (TB CHAMP study), therefore, can be used in children and adolescents. Testing for TBI among children is as for the national guidelines for DSTB. There is no requirement to test for TBI among children below 5 years.

According to current studies there is no evidence that FQs cause retardation of cartilage development in children (Hampel et al, 1997; Warren et al, 1997).

Pregnancy and breastfeeding:

This requires a risk – benefit assessment by the team treating the individual. Informed choice by pregnant woman on whether to take TPT or to defer TPT to the end of pregnancy.

Factors to consider;

4.) Time of pregnancy (first trimester versus later pregnancy)
5.) Pregnancy increases the risk of progression from infection to disease & risk of poor maternal and fetal outcomes should TB disease occur
6.) MDR/RR-TB in pregnancy is a serious condition with some of the drugs used being toxic to the fetus.

In a meta-analysis of observational studies including 2800 pregnant women who were prescribed fluoroquinolones for any indication, found no difference in the incidence of birth defects, spontaneous abortion or prematurity from that in unexposed pregnant women (Acar et al, 2019). Breast feeding is not a contra indication for Lfx prophylaxis as the concentrations of levofloxacin in breastmilk appeared to be far lower than the dose for infants, therefore unlikely to cause adverse effects in breastfed infants (National Institute of Child Health and Human Development; 2006). Though the observed effects of FQs on bone and cartilages in animals are been seen in humans, there is concern on prolonged use of fluoroquinolones in humans (Food and Drug Administration, 2018; European Medicines Agency; 2019; Medicines and Healthcare Products Regulatory Agency; 2024).

HIV infection: Levofloxacin can be used in people with HIV. No specific drug–drug interaction with ART has been observed in people with HIV exposed to MDR/RR-TB, and there is no need to test for infection before starting Lfx.

Contraindication: Lfx should not be given to people who are allergic to FQ, who have another contraindication to the same class of drugs or when there is potential drug–drug interaction. Lfx should be discontinued if the person develops a serious or severe adverse drug reaction.

Implementation considerations

In providing Lfx, factors such as age, risk of toxicity or interaction, co-morbidity, the susceptibility to drugs of the strain of the most likely source case, background resistance to FQs in MDR/RR-TB strains, availability and the individual's preferences have to be considered.

A paediatric formulation of Lfx can be used. For dosage refer WHO operational handbook on tuberculosis, 2024, page 47). If FQs cannot be used because of intolerance or resistance in the strain from the presumed source case, treatment with the other TB drugs used in some studies may be considered (e.g., E,Eto), although the evidence for their efficacy is much less certain (Schaaf et al, 2002; Baker et al, 2023).

Table 8: Features of preventive treatment regimen

Characteristic	**6H**	**4R**	**6Lfx**
Drug(s)	H	R	Lfx
Duration (months)	6	4	6
Frequency	Daily	Daily	Daily
Total no. of doses	182	120	182
Pill burden per dose (total per regimen), person weighing 50 kg using adult formulations (300mg)	1 (182)	3 (360)	1 (182)
Children	All ages; child-friendly (dispersible) formulation will be available for children with HIV on LPV/r or	All ages; no child-friendly formulation available, not generally feasible for children < 25 kg	All ages; child-friendly (dispersible) formulation will be available

	NVP		
Pregnant women	Safe for use	Advice avoid	Advice avoid unless indicated by risk benefit assessment by clinicians
Interaction with ART	No restriction	Advice avoid (Contraindicated: All PIs, NVP, doravirine and etravirine, TAF Adjust dose: DTG, RAL Use: TDF, EFV)	No restriction (may interfere with lamivudine clearance)
Toxicity	Hepatotoxicity (more), peripheral neuropathy, rash, gastrointestinal upset	Rash, gastrointestinal upset, hepatotoxicity (less), hypoprothrombinemia, orange discoloration of body fluids	Diarrhoea, nausea and bloating, arthralgia, inflamed or torn tendons, muscle pain or weakness, prolonged QTc interval, mood or behaviour changes, insomnia
Absorption	Best absorbed on an empty stomach; up to 50% reduction in peak concentration with a fatty meal		Absorption is not influenced by food. Concomitant steroid use may increase risk of tendon rupture. Multivalent cation-containing products including antacids (may contain aluminium), mineral supplements (e.g. iron or magnesium) or multivitamins may decrease absorption. Effect of warfarin may be enhanced.

Monitoring of treatment

Before initiating TPT, it is important to assess the possibility of side effects as well as patient compliance factors.

- The prophylactic treatment must be closely monitored until completion, at least on monthly intervals.
- Side effects should be monitored and if any, should be reported to the NPTCCD.
- These incidences to be presented and reviewed during the national level aDSM review meetings conducted by NPTCCD.

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- WHO operational handbook on tuberculosis. Module 1: prevention - tuberculosis preventive treatment, second edition. Geneva: World Health Organization; 2024. Licence: CC BY-NC-SA 3.0 IGO.

Chapter 8: PROCURES TO FOLLOW FOR FOREIGN NATIONALS DIAGNOSED OF DRTB

Sri Lanka is placed in close proximity to high burden MDR TB countries. With the frequent travelling between the countries, and foreign nationals coming to Sri Lanka for employment, there is a risk of getting them diagnosed as MDR TB while in Sri Lanka.

If a foreign national is diagnosed of MDR TB, following procedure is recommended.

1. Register them with separate MDR TB number as they also have to be counted for issuing second line drugs.

e.g., MDRTB/2024/FN/ 001

However, they will not be considered in developing national data, which will consider only the Sri Lankan nationals.

2. Notification and contact screening have to be carried out as per the standard procedure for the contacts within Sri Lanka.
3. Drug issuing – Second Line Drugs will be issued for the registration number provided for foreign nationals for the period they are residing in Sri Lanka and additional two weeks stock in case of they are returning to mother country.
4. The national TB programme focal point of the foreign country to be notified via email with the details of the patient.
5. Follow up of such patients will be as per the normal follow up.
6. Quarantine & travelling – If such a patient has to travel by air or sea, delaying it till sputum conversion (in bac. Confirmed PTB patients) or clinical improvement in other forms of TB is advised.

   The infection prevention control measures including wearing masks has to be adhered to.

Chapter 9: PROCEDURE IN THE EVENT OF A DEATH

Mycobacterium tuberculosis (MTB) is an airborne pathogen classified under biological hazard group 3, primarily transmitted via aerosols and occasionally through accidental inoculation. There is a risk of MTB transmission from cadavers to healthcare workers and others handling them, as MTB can remain viable in lung tissues for up to 36 days post-mortem. Case reports have documented transmission of MTB from cadavers to embalmers, highlighting the importance of strict infection control measures when handling cadavers of confirmed or suspected TB patients.

Healthcare workers involved in autopsies and embalmers are at high risk of acquiring tuberculosis from cadavers. However, the WHO does not classify TB as an infectious disease requiring bagging, and activities such as viewing, embalming, cosmetic enhancement, and hygienic preparation are not prohibited.

Those handling the bodies of suspected or confirmed TB patients should adhere to standard and airborne precautions. Autopsies on such bodies pose a high risk for MTB transmission, especially during aerosol-generating procedures like median sternotomy. Special airborne precautions are necessary, including using autopsy rooms with negative pressure and exhausting air outside the building. Air-cleaning technologies like HEPA filtration or UVGI can increase the number of air changes per hour (ACH). If these are not available, local exhaust ventilation should be used to reduce exposure to infectious aerosols. Healthcare workers performing autopsies should wear at least N95 disposable respirators or equivalents.

After an autopsy on a body with suspected or confirmed TB, sufficient time should be allowed for the removal of MTB-contaminated air before another procedure is performed in the same room. If this is not feasible, staff should continue wearing respirators while in the room.

Similarly, tissue or organ removal in embalming rooms from bodies with suspected or confirmed TB poses a high risk for MTB transmission, particularly during aerosol-generating procedures. Negative pressure ventilation should be used, and air exhausted outside the building. Air-cleaning technologies can increase ACH, and local exhaust ventilation should be considered to reduce exposure to infectious aerosols and embalming fluid vapors. Healthcare workers should wear at least N95 disposable respirators during these procedures. After tissue or organ removal, adequate time should be allowed for air clearance, or staff should continue wearing respirators if immediate reuse of the room is necessary.

All DRTB deaths have to be investigated at the district level with a panel consisting of CRP, DTCO and other relevant stakeholders.

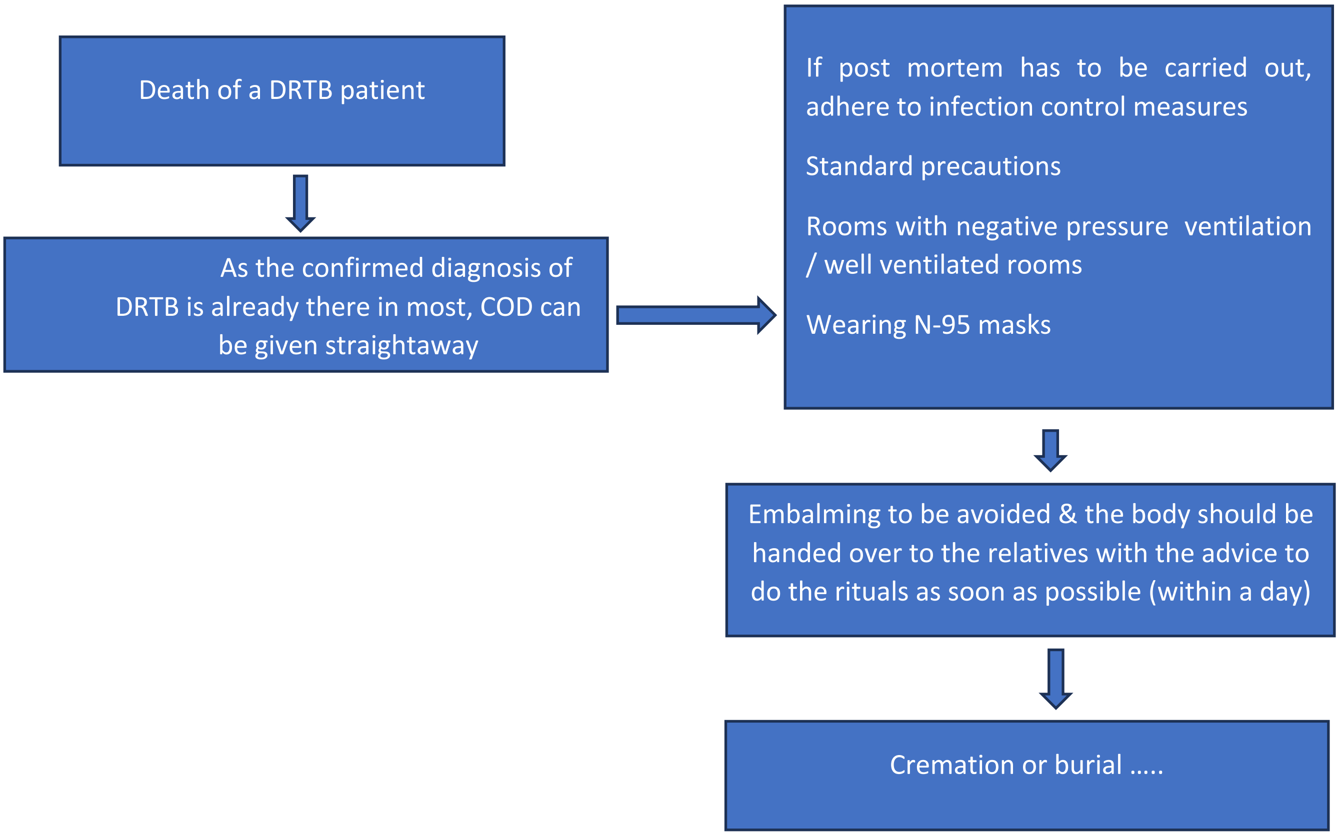

Figure 5:Flow chart to follow in the event of a DRTB death (MDR TB/ RR TB, Pre XDR TB, XDR TB)

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Chapter 10: MONITORING AND EVALUATION OF PMDT ACTIVITIES

With the introduction of the new 6month treatment regimen and the new case registration categories, the existing reports to be sent to the national programme are edited. Further, PMDT register is edited to include all the important details of the patient. A database of this PMDT register is expected to be kept at each District chest Clinic for all types of Drug-Resistant TB patients.

1. DRTB 1: Presumptive MDR register (Annexure 2)
2. DRTB 2: Presumptive TB screening summary (Annexure 3)
3. DRTB 3: Quarterly Report on DR-TB Case Finding (Annexure 4)
4. DRTB 4: Six-month interim report on the culture results of DR-TB patients registered 6-9 months earlier (To be filled after 9 months) (Annexure 5)
5. DRTB 5: Quarterly Report on treatment success of DR-TB patients [To be filled for cohort 9-12 months before (BPaL, BPaLM, OSSLR)] (Annexure 6)
6. DRTB 6: Quarterly Report on treatment success of DR-TB patients (Annexure 7)
7. DRTB 7: Side Effect Monitoring and Management during DR-TB Treatment (This form should be sent along with SE/SAE forms during reporting of adverse events to national level) (Annexure 8)
8. DRTB 8: Adverse Event Monitoring form (Annexure 9)
9. DRTB 9: Serious Adverse Event Monitoring form (Annexure 10)
10. DRTB 10: Death investigation form (Annexure 11)

Annexure 1: WHO recommendations on drug regimen – 2025

Table B. List of recommendations in the 2025 edition, where (a) is a new recommendation based on review of the new evidence and (b) is a reprinted recommendation where no new evidence was available or searched for the review

1. Treatment of drug-resistant TB using 6-month regimens	
1.1	**The 6-month bedaquiline, pretomanid, linezolid and moxifloxacin (BPaLM) regimen for MDR/RR-TB and pre-XDR-TB (b)** WHO suggests the use of the 6-month treatment regimen composed of bedaquiline, pretomanid, linezolid (600 mg) and moxifloxacin (BPaLM) rather than 9-month or longer (18-month) regimens in MDR/RR-TB patients. *(Conditional recommendation, very low certainty of evidence)*
1.2	**The 6-month bedaquiline, delamanid, linezolid, levofloxacin and clofazimine (BDLLfxC) regimen (a)** WHO suggests the use of a 6-month treatment regimen composed of bedaquiline, delamanid, linezolid (600 mg), levofloxacin, and clofazimine (BDLLfxC) in MDR/RR-TB patients with or without fluoroquinolone resistance. *(Conditional recommendation, very low certainty of evidence)*
2. Treatment of drug-resistant TB using 9-month regimens	
2.1	**The 9-month all-oral regimen for MDR/RR-TB (b)** WHO suggests the use of the 9-month all-oral regimen rather than longer (18-month) regimens in patients with MDR/RR-TB and in whom resistance to fluoroquinolones has been excluded. *(Conditional recommendation, very low certainty of evidence)*
2.2	**The modified 9-month all-oral regimens for MDR/RR-TB (a)** WHO suggests using the 9-month all-oral regimens (**BLMZ, BLLfxCZ and BDLLfxZ**) over currently recommended longer (>18 months) regimens in patients with MDR/RR-TB and in whom resistance to fluoroquinolones has been excluded. Amongst these regimens, using BLMZ is suggested over using BLLfxCZ, and BLLfxCZ is suggested over BDLLfxZ. *(Conditional recommendation, very low certainty of evidence)*
2.3	WHO suggests against using 9-month DCLLfxZ or DCMZ regimens compared with currently recommended longer (>18 months) regimens in patients with fluoroquinolone-susceptible MDR/RR-TB. *(Conditional recommendation, very low certainty of evidence)*
3. Treatment of drug-resistant TB using longer regimens (b)	
3.1	In multidrug- or rifampicin-resistant tuberculosis (MDR/RR-TB) patients on longer regimens, all three Group A agents and at least one Group B agent should be included to ensure that treatment starts with at least four TB agents likely to be effective, and that at least three agents are included for the rest of the treatment if bedaquiline is stopped. If only one or two Group A agents are used, both Group B agents are to be included. If the regimen cannot be composed with agents from Groups A and B alone, Group C agents are added to complete it. *(Conditional recommendation, very low certainty of evidence)*

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3.2	**Kanamycin and capreomycin** are not to be included in the treatment of MDR/RR-TB patients on longer regimens. *(Conditional recommendation, very low certainty of evidence)*
3.3	**Levofloxacin or moxifloxacin** should be included in the treatment of MDR/RR-TB patients on longer regimens. *(Strong recommendation, moderate certainty of evidence)*
3.4	**Bedaquiline** should be included in longer multidrug-resistant TB (MDR-TB) regimens for patients aged 18 years or more. *(Strong recommendation, moderate certainty of evidence)* **Bedaquiline** may also be included in longer MDR-TB regimens for patients aged 6–17 years. *(Conditional recommendation, very low certainty of evidence)* In children with MDR/RR-TB aged below 6 years, an all-oral treatment regimen containing **bedaquiline** may be used. *(Conditional recommendation, very low certainty of evidence)*
3.5	**Linezolid** should be included in the treatment of MDR/RR-TB patients on longer regimens. *(Strong recommendation, moderate certainty of evidence)*
3.6	**Clofazimine and cycloserine or terizidone** may be included in the treatment of MDR/RR-TB patients on longer regimens. *(Conditional recommendation, very low certainty of evidence)*
3.7	**Ethambutol** may be included in the treatment of MDR/RR-TB patients on longer regimens. *(Conditional recommendation, very low certainty of evidence)*
3.8	**Delamanid** may be included in the treatment of MDR/RR-TB patients aged 3 years or more on longer regimens. *(Conditional recommendation, moderate certainty of evidence)* In children with MDR/RR-TB aged below 3 years **delamanid** may be used as part of longer regimens. *(Conditional recommendation, very low certainty of evidence)*
3.9	**Pyrazinamide** may be included in the treatment of MDR/RR-TB patients on longer regimens. *(Conditional recommendation, very low certainty of evidence)*
3.10	**Imipenem–cilastatin or meropenem** may be included in the treatment of MDR/RR-TB patients on longer regimens. *(Conditional recommendation, very low certainty of evidence)*[20]
3.11	**Amikacin** may be included in the treatment of MDR/RR-TB patients aged 18 years or more on longer regimens when susceptibility has been demonstrated and adequate measures to monitor for adverse reactions can be ensured. If amikacin is not available, streptomycin may replace amikacin under the same conditions. *(Conditional recommendation, very low certainty in the estimates of effect)*

[20] Imipenem–cilastatin and meropenem are administered with clavulanic acid, which is available only in formulations combined with amoxicillin. Amoxicillin–clavulanic acid is not counted as an additional effective TB agent, and it should not be used without imipenem–cilastatin or meropenem.

3.12 **Ethionamide or prothionamide** may be included in the treatment of MDR/RR-TB patients on longer regimens only if bedaquiline, linezolid, clofazimine or delamanid are not used, or if better options to compose a regimen are not possible.

(Conditional recommendation against use, very low certainty of evidence)

3.13 ***P*-aminosalicylic acid** may be included in the treatment of MDR/RR-TB patients on longer regimens only if bedaquiline, linezolid, clofazimine or delamanid are not used, or if better options to compose a regimen are not possible.

(Conditional recommendation against use, very low certainty of evidence)

3.14 **Clavulanic acid** should not be included in the treatment of MDR/RR-TB patients on longer regimens.

(Strong recommendation against use, low certainty of evidence)[20]

3.15 In MDR/RR-TB patients on longer regimens, a **total treatment duration of 18–20 months** is suggested for most patients; the duration may be modified according to the patient's response to therapy.

(Conditional recommendation, very low certainty of evidence)

3.16 In MDR/RR-TB patients on longer regimens, a **treatment duration of 15–17 months after culture conversion** is suggested for most patients; the duration may be modified according to the patient's response to therapy.

(Conditional recommendation, very low certainty of evidence)

3.17 In MDR/RR-TB patients on longer regimens containing amikacin or streptomycin, **an intensive phase of 6–7 months** is suggested for most patients; the duration may be modified according to the patient's response to therapy.

(Conditional recommendation, very low certainty of evidence)

4. Regimen for rifampicin-susceptible and isoniazid-resistant TB (b)

4.1 In patients with confirmed rifampicin-susceptible, isoniazid-resistant tuberculosis, treatment with rifampicin, ethambutol, pyrazinamide and levofloxacin is recommended for a duration of 6 months.

(Conditional recommendation, very low certainty in the estimates of effect)

4.2 In patients with confirmed rifampicin-susceptible, isoniazid-resistant tuberculosis, it is not recommended to add streptomycin or other injectable agents to the treatment regimen.

(Conditional recommendation, very low certainty of evidence)

5. Monitoring patient response to MDR/RR-TB treatment using culture (b)

5.1 In multidrug- or rifampicin-resistant tuberculosis (MDR/RR-TB) patients on longer regimens, the performance of sputum culture in addition to sputum smear microscopy is recommended to monitor treatment response. It is desirable for sputum culture to be repeated at monthly intervals.

(Strong recommendation, moderate certainty in the estimates of test accuracy)

6. Starting ART in patients on MDR/RR-TB regimens (b)

6.1 Antiretroviral therapy is recommended for all patients with HIV and drug-resistant tuberculosis requiring second-line antituberculosis drugs, irrespective of CD4 cell count, as early as possible (within the first 8 weeks) following initiation of antituberculosis treatment.

(Strong recommendation, very low certainty of evidence)

7. Surgery for patients on MDR/RR-TB treatment (b)	
7.1	In patients with rifampicin-resistant tuberculosis (RR-TB) or multidrug-resistant TB (MDR-TB), elective partial lung resection (lobectomy or wedge resection) may be used alongside a recommended MDR-TB regimen. *(Conditional recommendation, very low certainty of evidence)*
8. Hepatitis C virus (HCV) and MDR/RR-TB treatment co-administration (a)	
8.1	In patients with MDR/RR-TB and HCV co-infection, the WHO suggests the co-administration of HCV and TB treatment over delaying HCV treatment until after treatment of MDR/RR-TB is completed. *(Conditional recommendation, very low certainty of evidence)*

Recommendation 1.2 The 6-month bedaquiline, delamanid, linezolid, levofloxacin and clofazimine (BDLLfxC) regimen

No.	Recommendation (NEW)
1.2	WHO suggests the use of a 6-month treatment regimen composed of bedaquiline, delamanid, linezolid (600 mg), levofloxacin, and clofazimine (BDLLfxC) in MDR/RR-TB patients with or without fluoroquinolone resistance. *(Conditional recommendation, very low certainty of evidence)*

Remarks

1. Drug susceptibility testing (DST) for fluoroquinolones is strongly encouraged in people with MDR/RR-TB. Although it should not delay the initiation of the BDLLfxC, the test results should guide the decision on whether levofloxacin or clofazimine should be retained.
2. This recommendation applies to the following:
   a. People with MDR/RR-TB or pre-XDR-TB (i.e. MDR/RR-TB and resistance to fluoroquinolones).
   b. Patients with less than 1 month of previous exposure to bedaquiline, linezolid, delamanid or clofazimine. When exposure is greater than 1 month, these patients may still receive the regimen if resistance to the specific medicines involved in such exposure has been ruled out.
   c. People with diagnosed pulmonary TB of all ages, including children, adolescents, PLHIV, and pregnant and breastfeeding women.
   d. People with all forms of extrapulmonary TB except for TB involving the CNS, osteoarticular TB, or disseminated forms of TB with multiorgan involvement.
   e. Children and adolescents who do not have bacteriological confirmation of TB or resistance patterns but who do have a high likelihood of MDR/RR-TB (based on clinical signs and symptoms of TB, in combination with a history of contact with a patient with confirmed MDR/RR-TB).
3. When resistance to fluoroquinolones is unknown, the regimen should be started as BDLLfxC and then adjusted based on the DST results. In cases of quinolone susceptibility, clofazimine should be dropped and the regimen started or continued as bedaquiline, delamanid, linezolid and levofloxacin (BDLLfx). In cases of resistance to fluoroquinolones, levofloxacin should be dropped and the regimen started or continued as bedaquiline, delamanid, linezolid and clofazimine (BDLC).

Recommendation 2.1 The 9-month all-oral regimen for MDR/RR-TB

No.	Recommendation
2.1	WHO suggests the use of the 9-month all-oral regimen rather than longer (18-month) regimens in patients with MDR/RR-TB and in whom resistance to fluoroquinolones has been excluded. *(Conditional recommendation, very low certainty of evidence)*

Remarks

1. The 9-month all-oral regimen consists of bedaquiline (used for 6 months), in combination with levofloxacin/moxifloxacin, ethionamide, ethambutol, isoniazid (high-dose), pyrazinamide and clofazimine (for 4 months, with the possibility of extending to 6 months if the patient remains sputum smear positive at the end of 4 months), followed by treatment with levofloxacin/moxifloxacin, clofazimine, ethambutol and pyrazinamide (for 5 months). Ethionamide can be replaced by 2 months of linezolid (600 mg daily).
2. A 9-month regimen with linezolid instead of ethionamide may be used in pregnant women, unlike the regimen with ethionamide.
3. This recommendation applies to:
   a. people with MDR/RR-TB and without resistance to fluoroquinolones;
   b. patients without extensive TB disease[32] and without severe extrapulmonary TB;[33]
   c. patients with less than 1 month exposure to bedaquiline, fluoroquinolones, ethionamide, linezolid and clofazimine; when exposure is greater than 1 month, these patients may still receive this regimen if resistance to the specific medicines with such exposure has been ruled out;
   d. all people regardless of HIV status;
   e. children (and patients in other age groups) who do not have bacteriological confirmation of TB or resistance patterns but who do have a high likelihood of MDR/RR-TB (based on clinical signs and symptoms of TB, in combination with a history of contact with a patient with confirmed MDR/RR-TB).

Recommendations 2.2 and 2.3 The modified 9-month all-oral regimens for MDR/RR-TB

No.	Recommendation (NEW)
2.2	WHO suggests using the 9-month all-oral regimens (BLMZ, BLLfxCZ and BDLLfxZ) over currently recommended longer (>18 months) regimens in patients with MDR/RR-TB and in whom resistance to fluoroquinolones has been excluded. Among these regimens, using BLMZ is suggested over using BLLfxCZ, and BLLfxCZ is suggested over BDLLfxZ. *(Conditional recommendation, very low certainty of evidence)*
2.3	WHO suggests against using 9-month DCLLfxZ or DCMZ regimens compared with currently recommended longer (>18 months) regimens in patients with fluoroquinolone-susceptible MDR/RR-TB. *(Conditional recommendation, very low certainty of evidence)*

Remarks

1. The recommended modified 9-month all-oral regimens comprise bedaquiline, linezolid and pyrazinamide in different combinations with levofloxacin/moxifloxacin, clofazimine and delamanid.
2. This recommendation applies to the following:
    a. People with MDR/RR-TB and in whom resistance to fluoroquinolones has been excluded.
    b. People with diagnosed pulmonary TB, including children, adolescents, PLHIV, and pregnant and breastfeeding women.
    c. People with extensive TB disease and all forms of extrapulmonary TB except for TB involving the CNS, osteoarticular TB or disseminated forms of TB with multiorgan involvement.
    d. People with MDR/RR-TB and less than 1 month of previous exposure to any of the component medicines of the regimen (apart from pyrazinamide and fluoroquinolones). When exposure is greater than 1 month, these patients may still receive one of the regimens if resistance to the specific medicines with such exposure has been ruled out.
    e. Children and adolescents who do not have bacteriological confirmation of TB or resistance patterns but do have a high likelihood of MDR/RR-TB (based on clinical signs and symptoms of TB, in combination with a history of contact with a patient with confirmed MDR/RR-TB).

Recommendations 3.1–3.17 Longer regimens

No.	Recommendation
3.1	In multidrug- or rifampicin-resistant tuberculosis (MDR/RR-TB) patients on longer regimens, all three Group A agents and at least one Group B agent should be included to ensure that treatment starts with at least four TB agents likely to be effective, and that at least three agents are included for the rest of the treatment if bedaquiline is stopped. If only one or two Group A agents are used, both Group B agents are to be included. If the regimen cannot be composed with agents from Groups A and B alone, Group C agents are added to complete it. *(Conditional recommendation, very low certainty of evidence)*
3.2	**Kanamycin and capreomycin** are not to be included in the treatment of MDR/RR-TB patients on longer regimens. *(Conditional recommendation, very low certainty of evidence)*
3.3	**Levofloxacin or moxifloxacin** should be included in the treatment of MDR/RR-TB patients on longer regimens. *(Strong recommendation, moderate certainty of evidence)*
3.4	**Bedaquiline** should be included in longer multidrug-resistant TB (MDR-TB) regimens for patients aged 18 years or more. *(Strong recommendation, moderate certainty of evidence)* **Bedaquiline** may also be included in longer MDR-TB regimens for patients aged 6–17 years. *(Conditional recommendation, very low certainty of evidence)* In children with MDR/RR-TB aged below 6 years, an all-oral treatment regimen containing **bedaquiline** may be used. *(Conditional recommendation, very low certainty of evidence)*
3.5	**Linezolid** should be included in the treatment of MDR/RR-TB patients on longer regimens. *(Strong recommendation, moderate certainty of evidence)*
3.6	**Clofazimine and cycloserine or terizidone** may be included in the treatment of MDR/RR-TB patients on longer regimens. *(Conditional recommendation, very low certainty of evidence)*
3.7	**Ethambutol** may be included in the treatment of MDR/RR-TB patients on longer regimens. *(Conditional recommendation, very low certainty of evidence)*
3.8	**Delamanid** may be included in the treatment of MDR/RR-TB patients aged 3 years or more on longer regimens. *(Conditional recommendation, moderate certainty of evidence)* In children with MDR/RR-TB aged below 3 years **delamanid** may be used as part of longer regimens. *(Conditional recommendation, very low certainty of evidence)*
3.9	**Pyrazinamide** may be included in the treatment of MDR/RR-TB patients on longer regimens. *(Conditional recommendation, very low certainty of evidence)*

No.	Recommendation
3.10	**Imipenem–cilastatin or meropenem** may be included in the treatment of MDR/RR-TB patients on longer regimens. *(Conditional recommendation, very low certainty of evidence)*[37]
3.11	**Amikacin** may be included in the treatment of MDR/RR-TB patients aged 18 years or more on longer regimens when susceptibility has been demonstrated and adequate measures to monitor for adverse reactions can be ensured. If amikacin is not available, streptomycin may replace amikacin under the same conditions. *(Conditional recommendation, very low certainty of evidence)*
3.12	**Ethionamide or prothionamide** may be included in the treatment of MDR/RR-TB patients on longer regimens only if bedaquiline, linezolid, clofazimine or delamanid are not used, or if better options to compose a regimen are not possible. *(Conditional recommendation against use, very low certainty of evidence)*
3.13	***P*-aminosalicylic acid** may be included in the treatment of MDR/RR-TB patients on longer regimens only if bedaquiline, linezolid, clofazimine or delamanid are not used, or if better options to compose a regimen are not possible. *(Conditional recommendation against use, very low certainty of evidence)*
3.14	**Clavulanic acid** should not be included in the treatment of MDR/RR-TB patients on longer regimens. *(Strong recommendation against use, low certainty of evidence)*[37]
3.15	In MDR/RR-TB patients on longer regimens, a **total treatment duration of 18–20 months** is suggested for most patients; the duration may be modified according to the patient's response to therapy. *(Conditional recommendation, very low certainty of evidence)*
3.16	In MDR/RR-TB patients on longer regimens, a **treatment duration of 15–17 months after culture conversion** is suggested for most patients; the duration may be modified according to the patient's response to therapy. *(Conditional recommendation, very low certainty of evidence)*
3.17	In MDR/RR-TB patients on longer regimens containing amikacin or streptomycin, **an intensive phase of 6–7 months** is suggested for most patients; the duration may be modified according to the patient's response to therapy. *(Conditional recommendation, very low certainty of evidence)*

Table 3.1 gives details of the grouping of medicines recommended for use in longer MDR-TB regimens; the groups are summarized here for clarity:

- Group A = levofloxacin or moxifloxacin, bedaquiline and linezolid;
- Group B = clofazimine, and cycloserine or terizidone; and
- Group C = ethambutol, delamanid, pyrazinamide, imipenem–cilastatin or meropenem, amikacin (or streptomycin), ethionamide or prothionamide, and *p*-aminosalicylic acid.

Annexure 2: DRTB 1: Presumptive MDR register

Presumptive MDR TB register

Date	Presumptive MDR TB no. e.g., PMDR/2024/Col/0001	District TB no. (If available)	Name in full	Sex	Age	Address & contact no.	Presumptive MDR category (Use the letter from the list)	Results Mention with type of resistance (if detected) + (RR,H, MDR)			Remarks
								Date	Xpert MTB, Ultra, XDR	Culture Liquid, solid	

Presumptive MDR TB categories

A - Contacts of MDR-TB patients	**G**- Institutionalized patients e.g., :- prisoners, other settings like barracks
B – Previously treated patients (Relapses, 1st line regimen failures, loss to follow up)	**H**- Healthcare workers
C - Non-converters / delayed sputum conversion **C.1.** - New- Sputum smear +ve at 2nd / 3rd month **C.2.**- Previously treated - Sputum smear +ve at 3nd / 4rd month	I- Those who return from abroad with active TB
D - Patients with history of repeated treatment interruptions	**J**- Drug addicts
E- TB patients treated outside the NTP	**K**- Children
F- Patients with TB/HIV co-infection	

Annexure 3: DRTB 2: Presumptive MDR screening summary

Presumptive MDR screening summary															
Q...... 20......															
District		Patients registered during:		Year			Quarter					Date of completion of the report			
Name of the DTCO												Signature			
Presumptive MDR category		No. screened	No. positive for MTB	No. with resistance	Type of resistance					Method of diagnosis				No. tx started	Remarks
					RR	MDR	H	Pre XDR	XDR	Xpert MTB, Ultra, Truenat	Xpert MTB XDR	Culture & DST: Liquid	Culture & DST: Solid		
A - Contacts of MDR-TB patients															
B – Previously treated patients (Relapses, 1st line regimen failures, loss to follow up)															
C - Non-converters / delayed sputum conversion	C.1. - New- Sputum smear +ve at 2nd / 3rd month														
	C.2.- Previously treated - Sputum smear +ve at 3nd / 4rd month														
D - Patients with history of repeated treatment interruptions															
E- TB patients treated outside the NTP															
F- Patients with TB/HIV co-infection															
G- Institutionalized patients e.g., :- prisoners, other settings like barracks															
H- Healthcare workers															
I- Those who return from abroad with active TB															
J- Drug addicts															
K- Children															

Annexure 4: DRTB 3: Quarterly Report on DR-TB Case Finding

Quarterly Report on DR-TB Case Finding

District		Patients registered during:	Year		Quarter		Date of completion of the report	
Name of the DTCO							Signature	

Type of resistance	New		Non converter (Tx, Rtx)		Treatment after lost to follow-up (Tx, Rtx)		Treatment after failure of tx (Tx, Rtx)		Relapse (Tx, Rtx)		Tx after DR-TB tx		Other		HIV status			Total
															Positive	Negative	Unknown	
	M	F	M	F	M	F	M	F	M	F	M	F	M	F				
RR - TB																		
MDR-TB																		
Pre-XDR-TB																		
XDR-TB																		
Total																		
Hr-TB																		
Other resistance																		
Total																		

Type of regimen	< 15 years		≥ 15 years		Total
	M	F	M	F	
BPaLM					
BPaL					
Other shorter regimen					
OSSTR					
OLTR(MDR)					
Individualized regimen(MDR)					
Hr TB					
Pre XDR					
XDR					
Other resistance					

Annexure 5: DRTB 4: Six-month interim report on the culture results of DR-TB patients registered 6-9 months earlier (To be filled after 9 months)

Six-month interim report on the culture results of DR-TB patients registered 6-9 months earlier (To be filled after 9 months)

Patients registered during:	Year		Quarter	Date of completion of the report	
Name of the DTCO				Signature	

Total number of cases registered (still on treatment)	No. started on treatment in te quarter	Smear and culture results (of patients still on treatment)									No longer on treatment				
		Smear Negative			Smear Positive			Smear Not done			Died	Lost to Follow up	Failed	Not evaluated	Moved to Pre XDR/XDR
		Culture Negative	Culture Positive	Culture Unknown	Culture Negative	Culture Positive	Culture Unknown	Culture Positive	Culture Positive	Culture Unknown					
MDR TB															
Pre XDR TB															
XDR TB															

Comments: Transfer in: Transfer out:

Annexure 6: DRTB 5: Quarterly Report on treatment success of DR-TB patients [To be filled for cohort 9-12 months before (BPaL, BPaLM, OSSLR)]

Quarterly Report on treatment success of DR-TB patients									
To be filled for cohort 9-12 months before (BPaL, BPaLM, OSSLR)									
Patients registered during:	Quarter		Year			Date of completion of the report			
Name of the DTCO						Signature			
Category	Number registered	Cured	Treatment completed	Treatment failed	Died	Lost to follow-up	Moved to Pre - XDR, XDR	Not evaluated	Remarks
BPaLM									
BPaL									
Other shorter regimen									
OSSTR/9 month all oral									
Total									
Hr - TB									
Other resistance									
Total									
Comments: Transfer in:		Transfer out:							

Annexure 7: DRTB 6: Quarterly report on treatment success of DR-TB patients

Quarterly Report on treatment success of DR-TB patients									
To be filled for cohort 24 months before (including OLTR)									
Patients registered during:	Quarter		Year			Date of completion of the report			
Name of the DTCO						Signature			
Category	Number registered	Cured	Treatment completed	Treatment failed	Died	Lost to follow-up	Moved to Pre XDR, XDR	Not evaluated	Remarks
BPaLM									
BPaL									
Other shorter regimen									
OSSTR/9 - month all oral									
OLTR/LTR(MDR)									
OLTR (Pre- XDR)									
OLTR (XDR)									
Total									
Comments: Transfer in:	Transfer out:								

Annexure 8: Side effect monitoring and management during DRTB treatment

DRTB 7: Side Effect Monitoring and Management during DR-TB Treatment (This form should be sent along with SE/SAE forms during reporting of adverse events to national level)

NOTE: This form should be filled during each monthly follow up or whenever patient reports any side effect and should be part of patient file

Patient Name: **MDR TB registration #** **MDR TB Treatment Site:**

Month of Treatment	Date of SE reporting by patient	New Event in the past 30 days (Include all new events or changes in pre-existing conditions that began in the past 30 days)	Date of onset of SE	Name of the drug(s) that might have caused the side effect	Action taken, management and any adjustment in treatment regimen	Date resolved	Outcome*	Severity†	Seriousness‡	Remarks

Please use an additional sheet if extra space is required

*Outcome: **R1** (Recovered/ resolved), **R2** (Recovering/resolving), **S** (Recovered with sequelae), **N** (Not recovered/ not resolved), **D** (Died), **U** (Unknown).

Name of the person filling in this page and initial……………………………………………

†SEVERITY: **1** (Mild), **2** (Moderate), **3** (Severe)

‡SERIOUSNESS: **N** (Not serious), **H** (Hospitalization (caused or prolonged)), **P** (Permanent disability), **C** (Congenital abnormality), **D** (Death)

‡All serious adverse events (SAEs) to be reported per NPTCCD SAE reporting mechanism under aDSM

Additional Comments:

Annexure 9: DRTB 8: Adverse Event Monitoring form

DCC:	
Participant Name:	
Address:	
Contact phone number:	

Adverse Events	**Description**
Optic Neuritis	
Peripheral Neuropathy (PN)	
Myelosuppression	
QT Interval Prolongation	
Hepatotoxicity	
Other adverse events	

Severity	☐ Grade 1 ☐ Grade 2 ☐ Grade 3 ☐ Grade 4

Action taken on study drug	☐ Dose not changed ☐ Drug discontinued permanently ☐ Drug discontinued temporarily ☐ Unknown
Outcome of AE	☐ Resolved, No Sequelae ☐ Resolved with sequelae ☐ Not resolved ☐ Fatal ☐ Unknown
Date of AE outcome/subsidence	____/_____/_______
Causality assessment	☐ Certain ☐ Probable or Likely ☐ Possible ☐ Unlikely ☐ Not related ☐ Unassessable
Remarks	

Annexure 10: DRTB 9: Serious Adverse Event Monitoring form

DCC:	
Participant Name:	
Address:	
Contact phone number:	

Serious Adverse Events	
Did the participant have any SAE?	Yes ○ No
Serious Adverse events	Death Life-threatening Inpatient hospitalization or prolongation Congenital anomaly or birth defect Persistent or sig. disability or incapacity Important medical event
Death	
Date of death	____/_____/________
Cause of death	
Causality assessment	Certain Probable or Likely Possible Unlikely Not related Unassessable
Remarks	

Event	
Life Threatening	
In Patient Hospitalization / Prolongation	
Congenital anomaly or birth defect	
Persistent or Sig. Disability or Incapacity	
Important medical event	
Episodes	◯ 1 ◯ 2 ◯ 3
Date of onset	___/____/______
Underlying cause	
Severity	Grade 1 Grade 2 Grade 3 Grade 4
Action taken on study drug	Dose not changed Drug discontinued permanently Drug discontinued temporarily Unknown
Outcome of SAE	Resolved, No Sequelae Resolved with sequelae Not resolved Fatal Unknown
Date of SAE outcome / subsidence	___/____/______
Causality assessment	Certain Probable or Likely Possible Unlikely Not related Unassessable
Remarks	

Annexure 11: DRTB 10: Death investigation form

(Death investigation to be carried out by a team at least including the Consultant Respiratory Physician and District Tuberculosis Control Officer)

District		MOH area	
Team		Patient particulars-	
Name	Designation	Name	
		Sex	
		Age	
		Address	
		DRTB no.	
		Date diagnosed/registered	
		Date treatment started	

TB disease information		
Patient registration category		
DRTB category	RR/ MDR/ INH/ Pre XDR/ XDR/………………	
Site of TB		
Diagnostic investigations	Date	Result
Xpert/MTB rif result		
Xpert XDR result		
culture and DST result		
Other		
DRTB drug regimen	Drugs	Date started

If any change in drug regimen		
Change made	Reason	Date
DOT provider details		
DOT provision details		
Follow up investigations	Date	Result
Sputum smear		
Culture & DST		
CXR		
Other………………..		
Complications during treatment		
Anti TB drug related side effects during treatment		
Co-morbidities		

DM		*COPD*				
Chronic renal disease		*IHD*				
Chronic liver disease		*Ca Lung*				
Bronchial Asthma		*Other……………*				
Summary:						
Date of death						
Place of death						
Immediate cause of death						
Underlying cause of death						
Associated causes of death						
Conclusion	Death due to TB		Death not due to TB		Indeterminate	

Points discussed	Action to be taken	Responsible person

CRP:		DTCO:	
Name		Name	
Designation		Designation	
Signature		Signature	
Date		Date	

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